

Quantitative Spatial Economics for Policy Evaluation

Topic 0: Course context and roadmap

Gabriel M. Ahlfeldt

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Course convener

- ▶ CV
 - ▶ 2002-2006: Studies (Diplomvolkswirt), *Uni Freiburg, Carlos III Madrid, FU Berlin*
 - ▶ 2006-2009: PhD (Dr. rer. pol) an der *Universität Hamburg*
 - ▶ 2009-2024: Professor (Urban Economics), *London School of Economics*
 - ▶ Since 2024: Professor (Econometrics), *HU Berlin*
- ▶ Research focus: Urban/spatial economics
- ▶ Editor *Regional Science and Urban Economics*
- ▶ Winner of 2018 Frisch Medal (with Steve Redding, Daniel Sturm, Niko Wolf)
- ▶ Commissioned research for *Aldar Properties, Bundesamt für Kultur* (Schweiz), *IHK Darmstadt, Europäischer Rechnungshof, OECD, WRI, DfT* etc.

Quantitative Spatial Economics

- ▶ What is it about?
 - ▶ **Economy consists of locations connected** through trade/migration/commuting
 - ▶ Understand what determines **how economic activities are distributed across space**
 - ▶ Understand how distribution will change in **response to shocks and policies**
 - ▶ Combines **theoretical** models, **empirical** data, and quantitative methods
 - ▶ **General equilibrium**—Integrates land, labour, and goods markets
- ▶ Why is it relevant?
 - ▶ General-purpose policy tool to **simulate** general equilibrium **effects of policies**
 - ▶ *white-box* approach complements *black-box* approaches (*Machine Learning*)
 - ▶ with particular strengths in **out-of-sample prediction**
 - ▶ Penetrating development, environment, labour, trade, urban economics, etc.

Why quantitative spatial economics for policy evaluation?

- ▶ QSE complements reduced-form evidence in policy evaluation.
 - ▶ Policies induce **spatial reallocation** of workers, firms, and economic activity.
 - ▶ Responses propagate spatially through commuting, migration (and trade).
 - ▶ There are effectively **no untreated units** within the spatial economy.
 - ▶ QSE predicts effects on outcomes that are **not directly observable**.
 - ▶ The model constructs the counterfactual equilibrium that cannot be observed.
 - ▶ These can be highly policy-relevant, in particular, changes in **welfare**.
- ▶ **Increasingly relevant**
 - ▶ Canonical models well-studied and widely applied in academia
 - ▶ Becoming more widespread in applied research

From measurement to policy appraisal

1. **Define the model and its building blocks.**
 2. Calibrate or **estimate parameter values** (model guides identification strategy)
 3. **Invert local fundamentals** conditional on the parameters and observed data.
 4. Define the policy experiment by **changing fundamentals or other primitives**.
 5. **Solve the new equilibrium** while holding the parameter values fixed.
 6. Compute changes in local outcomes, aggregates, and welfare.
- ▶ The sequence makes assumptions transparent and enforces internal consistency.
 - ▶ Credible empirical evidence and careful validation remain essential.

The full QSE curriculum

- ▶ A **semester-long PhD course** at the Berlin School of Economics
 - ▶ Open to advanced MSc students at Humboldt University.
- ▶ Twelve linked topics develop increasingly rich spatial models.
 - ▶ From neoclassical spatial equilibrium to quantitative urban models.
 - ▶ From urban commuting to quantitative spatial models (including trade).
- ▶ The course combines **theory and computational application**.
 - ▶ Weekly problem sets and regular programming.
 - ▶ Numerical solution algorithms, replication directories, data, and toolkits.
- ▶ A **cumulative learning process**: later exercises build on earlier code.
 - ▶ The full format develops computational independence.
 - ▶ A one-week course necessarily uses a different learning model.

The focused CEMFI route

- ▶ We select five topics and follow **one coherent line**.
 - ▶ Topic 1: Rosen–Roback and the quantitative Glaeser–Gottlieb framework.
 - ▶ Topic 5: Discrete choice and preference heterogeneity.
 - ▶ Topic 6: The Ahlfeldt–Redding–Sturm–Wolf model.
 - ▶ Topic 7: Quantification of primitives and fundamentals.
 - ▶ Topic 8: Counterfactual policy evaluation.
- ▶ Lectures develop the theory; tutorials provide **guided computation**.
 - ▶ Build on the ARSW2015 toolkit; an accessible version of the replication directory with didactic counterfactuals, codebook, pseudo-code, etc. to develop intuition for computational implementation of the model

Lectures and practical sessions

Day	Lecture	Practical session (15:00–17:00)
Monday	Topic 1	—
Tuesday	Topic 5	—
Wednesday	Topic 6: model	ARSW2015 toolkit setup
Thursday	Topic 7: quantification	Reproduce quantification
Friday	Topic 8: counterfactuals	Reproduce counterfactuals

- ▶ Teaching-assistant sessions guide you through the code step by step.
 - ▶ The goal is to understand how **theoretical objects map into code**
 - ▶ Not merely to press *Run*.

Optional capstone: transport appraisal

- ▶ Construct travel-time matrices for **two Berlin transport networks**.
 - ▶ Compare the complete U5 with a network without the central connection.
- ▶ Translate the accessibility change into the ARSW model.
 - ▶ Solve for changes in employment, wages, rents, and welfare.
- ▶ The capstone connects **GIS and routing to spatial equilibrium**.
 - ▶ It is an **independent and optional** extension, but designed to be **doable**.
 - ▶ Combines two toolkits
 - ▶ ARSW2015 toolkit: quantify and simulate the spatial equilibrium model.
 - ▶ TTMATRIX toolkit: construct bilateral travel-time matrices.
- ▶ A chance for you to apply, test, and sharpen your skills...

What this week can—and cannot—deliver

► By Friday, you should be able to

- Read the equilibrium structure.
- Understand the quantification logic.
- Navigate the toolkit.
- Interpret counterfactual output critically.

► Beyond this week

- Write solvers from scratch.
- Develop new identification strategies.
- Build production-ready policy models.

Key is to understand how **primitives** (parameters and fundamentals) **map into endogenous outcomes** and how to **express policies as changes in primitives**

Policy → change in primitive → change in equilibrium outcome

Continue with the full experience

- ▶ Interested students can join the **semester-long course in Berlin**.
 - ▶ Visiting students work through the complete problem-set sequence.
 - ▶ They join the wider www.bqse.de quantitative spatial economics community.
- ▶ Information and applications:
 - ▶ **QSE visiting PhD students programme**.
- ▶ Participation is generally possible **without a course fee**.
 - ▶ Confirm practical arrangements and current availability in advance.